

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

29th November 2024

SP2 - Life Insurance

Time allowed: 3 Hours 15 Minutes (09.30 – 12.45 Hours)

Total Marks: 100

Q. 1) A life insurance company, ABC Ltd., has been primarily selling non-participating endowment assurance products since its establishment 30 years ago.

- a)** State the key factors that the life insurance company should consider when setting surrender values. (5)

Currently, the surrender values for the product vary by policy duration and are calculated as follows:

Surrender Value = Total Premium Paid to date * Max (Factor A, Factor B), where

Factor A is prescribed by the local regulatory body, and

Factor B is based on proxy asset share for determining surrender scales.

The insurance regulator has decided to increase Factor A with immediate effect, and the change shall be applicable for both new and existing policyholders.

A significant majority of the book for ABC Ltd. consists of policies with premium payment term 5 years and policy term 10 years. Following shall be the change in Factor A for these policies:

Duration/Year	1	2	3	4	5	6	7	8	9	10
Old Factor A	0%	30%	50%	70%	90%	100%	110%	120%	135%	150%
New Factor A	50%	60%	70%	80%	90%	100%	110%	120%	135%	150%

- b)** Evaluate these changes from the perspective of both customers and shareholders of ABC Ltd. (8)

- c)** The life insurance company is in the process of establishing its withdrawal assumption for the upcoming valuation year. Discuss the factors that the life insurance company should consider when changing the withdrawal assumptions for its endowment assurance product, following the regulatory change. (10)

- d)** Evaluate the influence on other assumptions emerging from secondary effects that may arise from this regulatory change. (6)

[29 Marks]

Q. 2) In Country of Indiana, life insurance policies often offer the option to purchase additional death benefit/ sum assured without requiring additional medical evidence, all while maintaining the original premium rates at the time of exercising the option.

- a)** Explain potential reasons for policyholders to opt for exercising this option. (2)

- b)** Discuss with an example a risk that the insurers in Indiana face due to offering such option. Also, enlist corresponding risk mitigation options that the insurers can deploy. (2)

Company X is currently in the process of developing a new product and is exploring the possibility of incorporating additional mortality options.

- c)** Suggest two mortality options (other than enhanced sum assured) that could be incorporated into the new product and possible reasons why a policyholder may choose to exercise these options. (4)

[8 Marks]

Q. 3) Country Lagoona has a unique and large retail life insurance market where a company is allowed to sell only one type of insurance policy. Companies ABC and XYZ sell unit-linked and without-profits endowment assurance products, respectively, with a fixed policy term of 15 years. While both the companies are market leaders in their respective lines of business, ABC currently has low level of free assets.

- a)** Stating the investment principles, compare and contrast the probable investment strategy for the two companies. (10)

- b) Following a change in international market practices, local insurance regulator has asked the insurers to assess capital requirements using a risk-based approach. Discuss the risks that these companies will hold capital for in the risk-based regime. (8)

[18 Marks]

Q. 4) The Chief Actuary of a large life insurance company has asked his team to perform the analysis of surplus arising in the current financial year.

- a) Describe, with practical examples, how the results of such surplus analysis would be used by the Chief Actuary. (11)

Note: Consider the implications on all aspects of actuarial work i.e., pricing, valuation, risk management, process and stakeholder management.

As an outcome of the analysis, it has been observed that expense surplus of the company is negative.

- b) Discuss what could be the possible reasons for such an outcome and list possible management actions that the company can take to avoid negative expense surplus in subsequent years. (6)

Chief Actuary has asked the Valuation actuary to reassess the expense assumptions to be used for valuation exercise.

- c) Explain the expense investigation that shall be carried out for setting the expense assumptions to be used for valuation exercise. (5)

[22 Marks]

Q. 5) A recently established company is in the process of preparing for the introduction of a new without-profit endowment assurance product.

- a) Discuss advantages and disadvantages for the company of choosing captioned product as first product offering. (10)

The company is planning to partner with a leading reinsurer which has been an active reinsurer of the respective geography for many years.

- b) List the reasons why a life insurance company may use reinsurance. (4)

- c) State the factors that the life insurance company should consider before determining the retention limit under its reinsurance arrangements. (5)

- d) Discuss the risks that the life insurance company is exposed to from the reinsurance arrangement. (4)

[23 Marks]
